

A MULTIPHYSICS APPROACH TO LFR ANALYSIS

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Abstract

A preliminary hexagonal geometry of the SMR model of the ALFRED (Advanced Lead Fast Reactor Demonstrator) concept design for a lead cooled reactor is analyzed with a computational platform that integrates different computational tools into the common framework given by the SALOME platform software. The purpose of the paper is to investigate the multiphysics (thermal-hydraulic-neutronic) approach to this SMR model by coupling the DRAGON-DONJON codes, for lattice calculations and full core simulation, with a 3D-porous media thermal-hydraulic code, FEMUS. In particular, the lattice code, DRAGON, is used to evaluate the macroscopic cross sections over the hexagonal lattice devised for ALFRED, collapsing the microscopic cross section data into 33 groups, parametrized on fuel and moderator temperature, density and power distribution. With the macroscopic cross sections for the various lattice cells of the reactor defined, the distribution of neutron flux in the core is obtained by the full core simulation with the DONJON code, that may be used in the simulation of fuel management, reactor operation or accident scenarios. In transient simulations (in a quasi-static approach), at each time step, the neutron flux is computed with the corresponding thermal source while the thermal-hydraulic module (FEMUS) computes the distribution of the coolant velocity, pressure and temperature fields in the reactor core. Then the neutron modules can receive all the feedback variables, as fuel temperature, moderator temperature and density that can be cast in input to the DRAGON code, where macroscopic cross sections are interpolated and total neutron fluxes evaluated. In this way, one can obtain an iterative process for studying the transient evolution of the model of the ALFRED fast lead cooled reactor as SMR case-study and preliminary results about his time-dependent behavior can be fully analyzed.

1. INTRODUCTION

In recent times, with challenges associated to anthropic activities environmental impact and global climate changes more and more evident, nuclear energy has been considered able to play a long-term role for meeting the world's increasing energy demand. An interest in small and simple units able to generate electricity from nuclear power is growing due to the need of reducing the impact of capital costs and provide power to larger and smart grid systems.

The International Atomic Energy Agency (IAEA) defines "Small Modular Reactors (SMRs) as nuclear reactors, generally 300 MWe equivalent or less, designed with modular technology using module factory fabrication, pursuing economies of series production and short construction times" [1]. The World Nuclear Association 2015 report on SMR standardization of licensing and harmonization of regulatory requirements, wrote that the enormous potential of SMRs is based on features such as small power and compact architecture that allow building of SMR units directly at factory, improving the level of construction quality and efficiency [1].

TABLE 1. ALFRED SMR core main parameters (reproduced from Ref. [3,4])

	Parameters	Unit	Value
PIN	Thermal Power	MW	300
	Pellet hollow diameter	mm	2.0
	Pellet radius	mm	4.5
	Gap thickness	mm	0.15
	Clad thickness	mm	0.6
	Pin diameter	mm	10.5
	Bottom plug length	mm	50
	Gas plenum height	mm	550
	Bottom insulator height	mm	10
	Active height	mm	600
	Upper insulator height	mm	10
	Spring length	mm	120
	Upper plug height	mm	50
Fuel Assembly	Lattice pitch (hexagonal)	mm	13.86
	Pins per FA	-	126
	Wrapper thickness	mm	4.08
	Distance between to wrappers	mm	5.0
	Average coolant velocity	m/s	1.28
CORE	Inner/Outer FAs number	-	56/78
	Inner Vessel Radius	m	1.475
	Cycle lenght	month	12
	Number of batches	-	5
FUEL	Pu vector	^{238}Pu	2.348
		^{239}Pu	57.015
		^{240}Pu	26.951
		^{241}Pu	6.069
		^{242}Pu	7.616
	U vector	^{234}U	0.003
		^{235}U	0.409
		^{236}U	0.010
		^{238}U	99.578
	Inner/Outer enrichment ($\text{Pu}+^{241}\text{Am}$)/($\text{Pu}+^{241}\text{Am}+\text{U}$)		atom% 20.5%/26.2%

Furthermore the small size and passive safety system could drive to massive use of SMRs in countries with small grids and less experience of nuclear power due to a more flexible financing policy. The lower power reduces the reactor radioactivity inventory, minimizes costs for a specific SMR design and during decommissioning, at the end of the core lifetime, the small size makes easier to remove reactor modules. There are different options for SMR's technology since these technologies allow the construction of a large range of small and simple reactors with long operational period before refueling.

This paper is devoted to the analysis of a fast neutron lead cooled reactor with a core in hexagonal geometry. In particularly the multi-physics approach, described in this paper, is focused on the conceptual design of ALFRED (Advanced Lead Fast Reactors European Demonstrator) [2,3,4], that is based on the sustainability of the nuclear energy source in the long term and on the high level of reactor safety design due to the intrinsic features of the lead coolant. The features of ALFRED in terms of thermal power and core size (small core sizes are more suitable for a joint neutronic-thermo-hydraulic modeling) makes it an interesting case-study that can be eventually extended to other SMR of the same class.

The details of the ALFRED core main parameters are reported in Table 1 [2,3,4] and a core configuration in Figure 1.

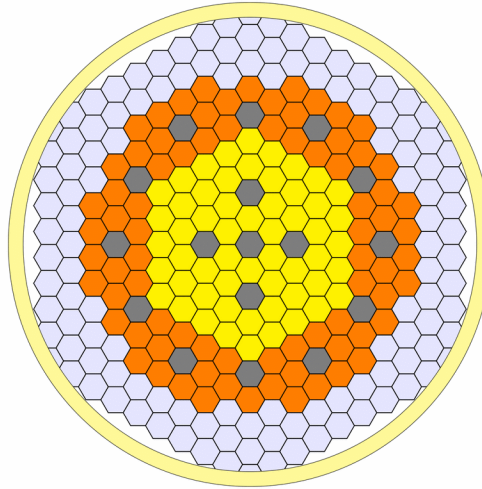


FIG. 1 Core geometry for ALFRED where inner MOX fuel with 20.5% in plutonium (yellow) and outer MOX fuel with 26.2% in plutonium (orange) (reproduced from Ref. [4]).

2. THE APPLICATION OF THE DRAGON LATTICE CODE.

The hexagonal geometry, as shown in Figure 2, has been built by following symmetry criteria in order to save computational time. The first step of a multi-scale neutronic approach consists of the creation of data structures containing macroscopic cross sections about the elements that form the nuclear reactor fuel.

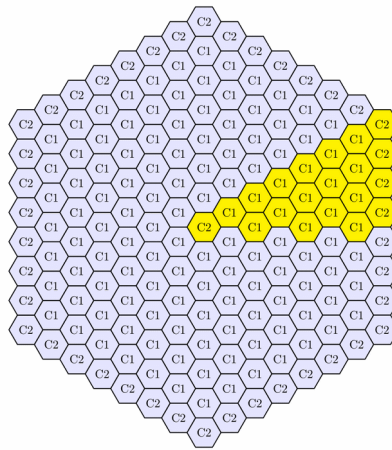


FIG. 2 Hexagonal geometry of the fuel assemblies used by Dragon-Donjon-FEMUS

This is obtained by using the lattice code DRAGON, version 5, released by Ecole Polytechnique of Montreal, a computer code designed around solution techniques of the neutron transport equation, which can simulate the neutronic behavior of a fuel assembly in a nuclear reactor [5]. Based on the work in [2,3,4] for the project data of ALFRED, we use three kind of macroscopic cross sections: two for fuel assemblies with different enrichment in plutonium and one for structural and control materials such control rods, safety rods and plenum lead. The lattice code DRAGON obviously starts from the microscopic cross sections libraries of nuclear data. In this case we used JEFF 3.1.2 cross section library with 315 energy groups (as reworked by Santamarina, Hebert and Hfayed) that, by setting nuclei properties, creates the needed micro-libraries.

Elementary nuclei are chosen taking into consideration fuel, coolant and cladding temperature together with nuclei density in order to consider Doppler and material expansion effects.

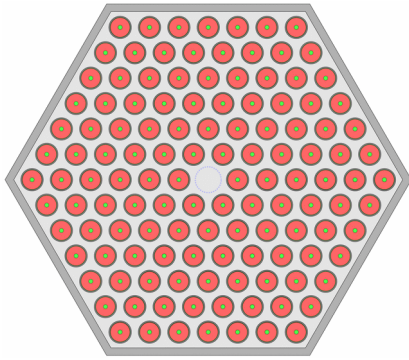


FIG. 3 Fuel assembly with 126 fuel pins, fuel pin (Ref. [4])

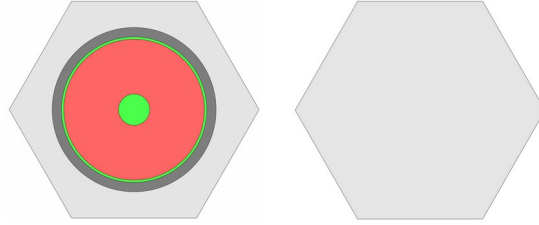


FIG. 4 Cell C1(left) composed by lead, helium gap, cladding and cell C2 (right) only composed by lead

The neutron transport equation in the lattice, is solved with the collision probability method [6] using a hexagonal geometry according to model geometry of the fuel assembly as defined in [2,3,4] (see Figures 3 and 4). The cell C1, shown in Figure 4, constitutes the fuel pin which is composed by lead, cladding, helium gap and nuclear fuel, while the cell C2 constitutes the structural parts of the fuel assembly. Then, the data structures of the macroscopic cross sections have been created using a series of calculations that take into account self-shielding, fuel burn-up, temperature and concentration changes. These data structures, called multi-compo, are condensed in 33 energy groups in order to reduce computation time while maintaining a high level of accuracy.

3. THE FULL CORE ANALYSIS (DONJON CODE)

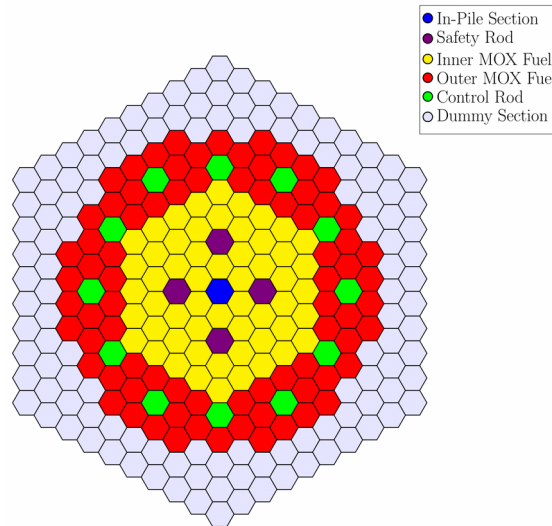


FIG. 5 Core map of reactor composed by assembly of In-Pile section (blue), Safety rods (violet), Control rods (green), Inner MOX Fuel (yellow), Outer MOX Fuel (red) and Dummy elements (gray) (Ref. [4])

The comprehensive neutronic behavior of this nuclear reactor is investigated with the full core simulation by using the DONJON, version 5, code and the group cross sections from the previous lattice code step in order to obtain the reactor characteristics as flux and power distribution. ALFRED (Advanced Lead-cooled Fast Reactor European Demonstrator) [2,3,4] is a demonstrator of the lead fast reactor technology, with a foreseen

thermal power of 300 MW and his composition of U and PU vectors is a typical MOX fuel but with enrichment different for inner (yellow FA) and outer (red FA) core zones, with the purpose of a more uniform (flat) flux distribution on the whole core. Some parameters of the reference model used in this work have been reported in Table 2 [3].

TABLE 2. Reference data used to model the ALFRED reactor (Ref. [4]).

Parameters	Value
Thermal Power [MW]	300
Total Fuel Assembly	134
Inner Fuel Assembly	56
Outer Fuel Assembly	78
Control Rods	12
Safety Rods	4
In-Pile Section	1
Inner vessel inner/outer radius [m]	1.475/1.525
Total Height Vessel [m]	3.50
Active Height [m]	0.60
Average Core Flow [m/s]	1.28
Coolant inlet/outlet Temperature [°C]	400/520

The core map has a hexagonal geometry as shown in Figure 5. The geometry of the whole reactor core has been considered as a hexagonal prism, as in Figure 5, comprising 10 rings and 16 floors with different level of accuracy, in particular in the core zones. The full core simulation employs finite elements methods to resolve neutron diffusion and simplified PN equations [7]. The fuel bundles have a distinct set of properties that are recovered from the reactor database, obtained from lattice calculations, and then interpolated according to the specified global or local parameters.

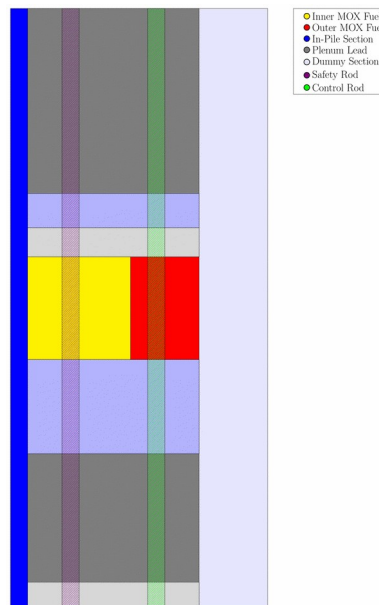


FIG.6 Axial profile of reactor used during the full core simulation.

In this way, it is possible to interpolate the values of fuel and coolant temperatures that are obtained from the thermal-hydraulic analysis, and set an iterative approach producing values for both neutronic and thermal-hydraulic codes. Flux distribution, power distribution, peak power factor and effective multiplicative factor are computed by full core simulation code. In order to validate the obtained computations, we take into account the reference multiplication factor and compare the results with other works [3]; in Table 3 we show the results

from computation Monte Carlo code, MCPN [9], considering BOC at 2 years and EOC at 3 years about the study in reference [3]. Therefore, in these works the cycle swing in one year (evaluated between BOC and EOL) is $\Delta k_{\text{eff}}/k_{\text{eff}} = -2580$ pcm for MCPN code and about -2500 pcm for ERANOS code [8]; these values are closed enough with DONJON code, where the burn-up swing is about -2300 pcm.

Anyway, the difference of k-effective between computational codes is wide. However the results are inside a reasonable range of values since this computational design is an ALFRED-like model and so some dimensional parameters are not up to date, some are a little different and others are not available.

TABLE 3. K-effective with control rods withdraw from different computational approaches. [3]

Time (days)	DONJON	MCNP
0	1.11403	1.0804
365	1.08531	1.0510
730 (BOC)	1.05929	1.0247
1095 (EOC)	1.03549	0.9988

4. THE NUMERICAL PLATFORM

The principal purpose for code coupling in the wrapping numerical platform is to couple the solution of a multi-physics and multi-scale three-dimensional problem inside a simplified and more comprehensive framework. The Salomé project facilitates the coupling of scientific mesh-based codes [10], in our case between the full core simulations by the DONJON code and the open source code FEMUS, thanks to its architecture and suite of tools that provide several data interfaces and exchange across the different codes.

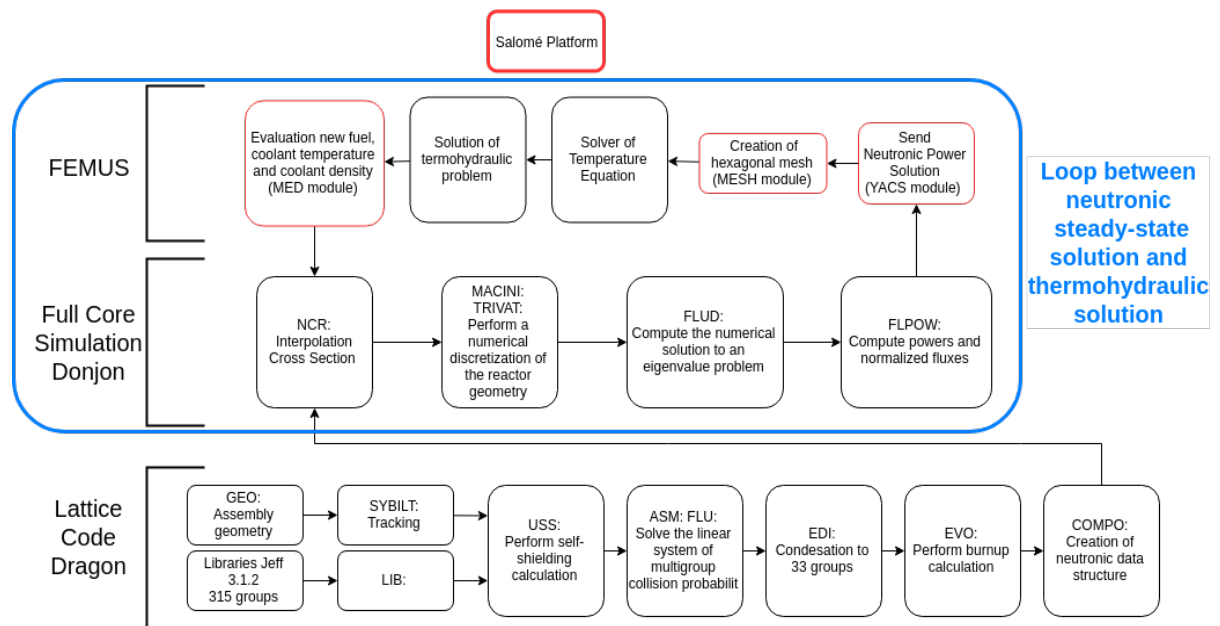


FIG. 7 Flowchart of the coupled neutronics-thermalhydraulics problem.

Inside Salomé there are libraries, called MED (Modelisation et Echanges de Donneés, i.e. Data Modelization and Exchanges) modules, that provide a library for storing and recovering computer data in a suitable format. The MED data transfer is based on the association of numerical meshes with fields and allows the data exchange between solvers and codes. The GEOM and MESH modules of the Salomé platform have the main function to draw and create meshes in MED format, respectively.

Other Salomé module such as the YACS module is a tool to supervise execution of complex interconnected scientific applications as object structure available during execution of DONJON code. The

integration of a code on the Salomé platform is obtained by generating an interface with functions available in the MEDMem library that allows a data transfer from the platform to the code and then from the code to the interface.

In addition to neutronic codes, previously described, CFD modules [11,12] have the function to solve energy and temperature equations starting from a given neutronic power density distributions. Temperature fields, evaluated with CFD codes, FEMUS or OPEN-FOAM, may be introduced into the DONJON neutronic code with the purpose of interpolating macroscopic cross sections according to the local temperature fields and local coolant (lead) density distributions. As result of this temperature distribution the neutron flux changes and defines a new power density distribution inside these CFD modules. In particular, the nuclear reactor is modeled as a *porous* medium where we consider the real assembly geometry and the effective mass flow: this approach allows us to consider that the lead cooling is effective only on a part of the assemblies, being the available cooling channels limited by fuel assemblies and structural materials. A flowchart that summarizes the behavior of this multi-physics approach with the Salomé platform is shown in Figure 7.

5. RESULTS AND CONCLUSION

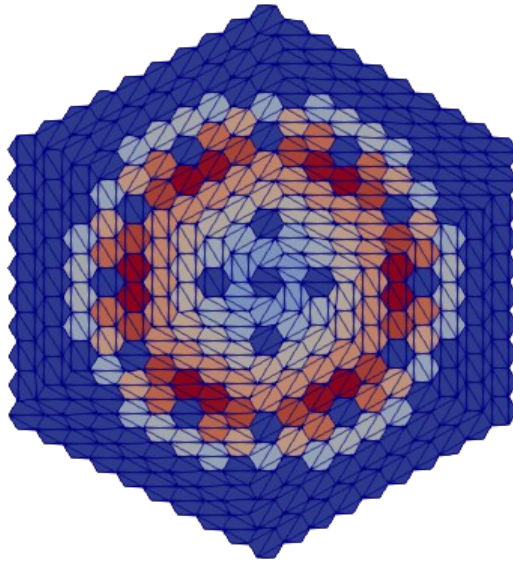


FIG. 8 Finite element solution for the neutronic power of reactor ALFRED with a mesh optimized for the neutron transport equation.

In both CFD and neutron transport codes, the various fields have been computed using a FE method over different refined meshes. The coarse mesh, as shown in Figure 8, consists of hexagonal assemblies. Each assembly element is divided in such a way that each subelements have quadrangular surfaces. This coarse mesh is then refined by using midpoint refinement algorithm several times. Each hexagonal element have constant properties but several field points.

In conclusion a collection of images for the flux distribution, power distribution and temperature field is reported in Figures 9-13, that describe the behavior of the core model on different sections at different heights.

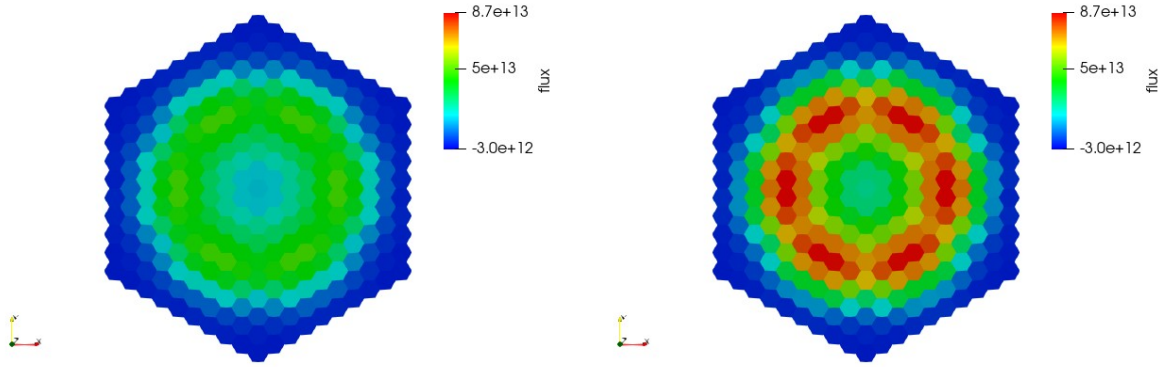


FIG. 9 Flux distribution over a section of core reactor at 150 cm (left) and 175 cm (right).

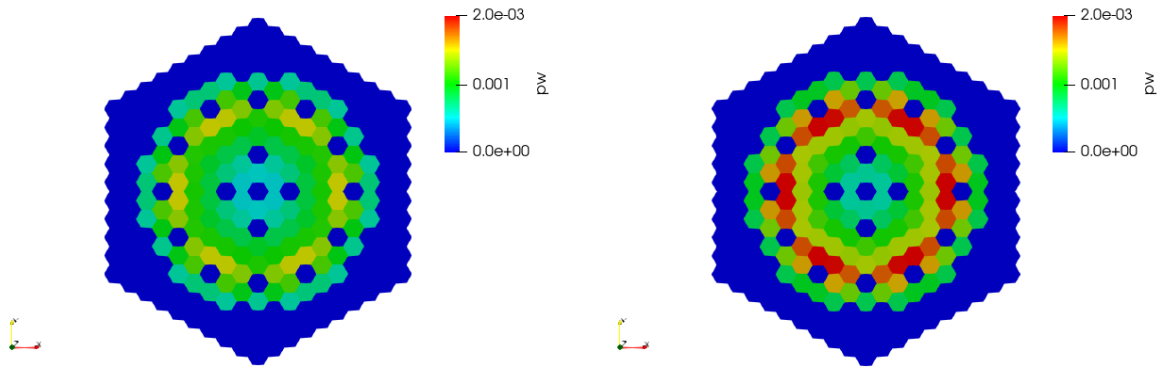


FIG. 10. Normalized power distribution over a section of core reactor at 150 cm (left) and 175 cm (right), i.e. at the top and at the middle of the core active zone.

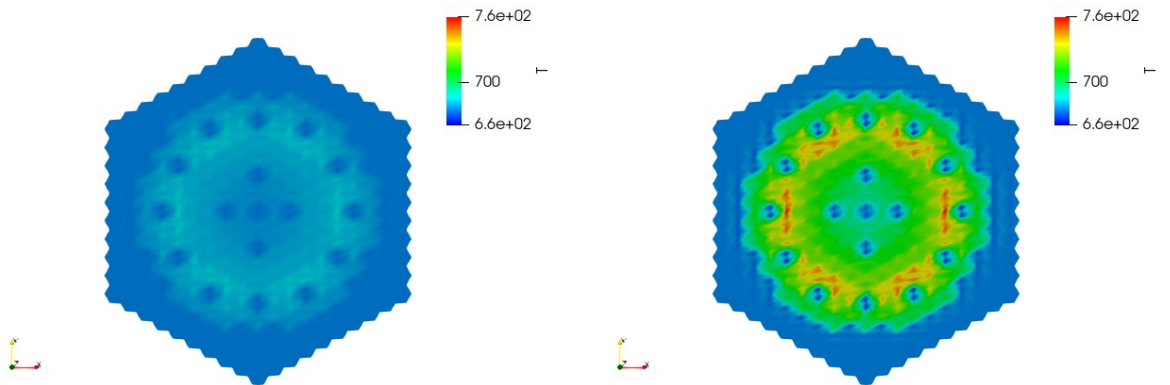


FIG. 11. Temperature field over a section of core reactor at 150 cm (left) and 175 cm (right).

In particular Figure 9 shows the total neutronic flux over two different planes of the reactor. On the left we have the distribution on the section at 150 cm while on the right the section is located at 175 cm from the bottom. It is important to recall that the neutron flux is computed using 33 groups and this is the integral over all energy values. Figure 10 shows, from the left to the right, the normalized power distribution over a section of core reactor at 150 cm and 175 cm, respectively. The thermal power is computed based on the peak factors

which represent the ratio between the assembly thermal source and the average assembly power. The thermal source contributes by advection thanks to the velocity field of each assembly and is diffused by solving the 3D energy equation of the porous media core model. Figure 11 shows the temperature over the same section of core reactor at the same heights as before. The temperature distribution resembles the thermal power distribution and shows cooled area in the assemblies where there is no heat generation.

In Figure 12 we show a summary for neutron flux, thermal power distribution and temperature field. The neutron flux, the power distribution and the temperature field are reported on the left, center and right over the sections of core reactor defined by the z-value of 150 cm (top), 175 cm (center) and 200 cm (bottom), respectively.

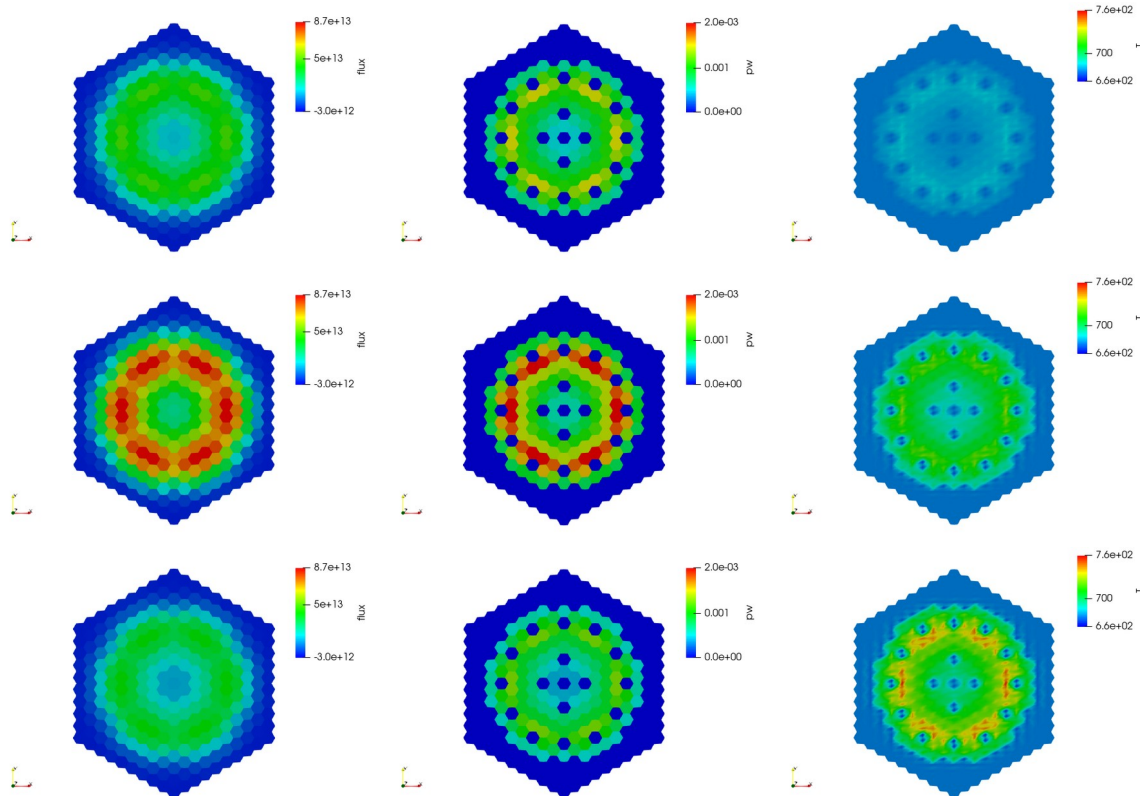


FIG. 12 Flux distribution (left), Power distribution (middle), Temperature field (right) at section of core reactor at 150 cm (top), 175 cm (center) and 200 cm (bottom).

The solution is obtained after several iterations between the thermal-hydraulic and the neutron code with temperature and density feedback corrections. In Figure 13 the neutron flux and the thermal power field over a vertical section of the reactor are shown. The neutron flux along y-axis is showed in Figure 14. The DONJON code can compute the neutron flux for different burn-up inventory and a large range of control rod movement. This allows a detailed analysis of the reactor and its large capability in term of fuel available from different models of this fourth generation fast reactor.

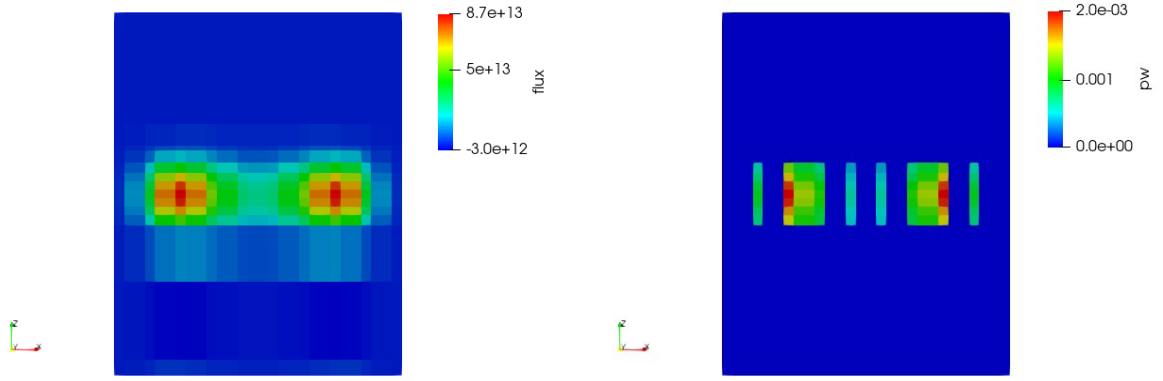


FIG. 13 Neutron flux and thermal power field over a vertical section of the reactor

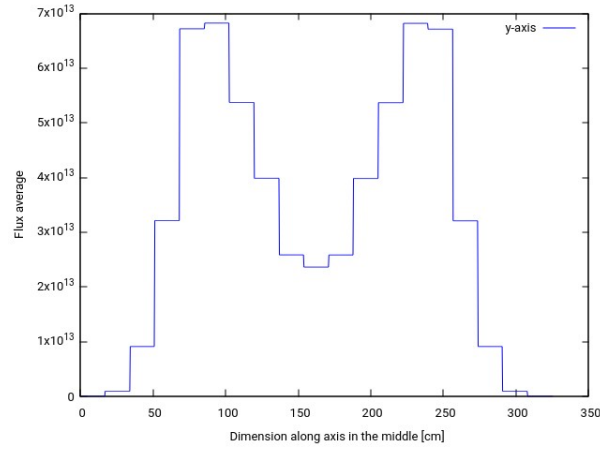


FIG. 14 The distribution of neutron flux average (piecewise constant in each cells) along y-axis

The results obtained in this simple case study, open a perspective of an extensive similar approach to other models: SMRs look as ideal candidates due to their size and compactness (mainly in terms of core dimensions in the FR case).

Finally this work need more time for refine data modeling about the model of ALFRED and has the aim to present the numerical platform for coupling the neutronic codes and thermal-hydraulic code; anyway we are sure that the numerical platform is a good help to analysis the physics behavior of nuclear reactor in project phase on terms of reliability.

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